BEATS: Audio Pre-Training with Acoustic Tokenizers

A self-supervised framework with iterative acoustic tokenizers and masked discrete label prediction

SOTA mAP 50.6% (AS-2M) | ESC-50 Acc 98.1% | Code: https://aka.ms/beats

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The authors report state-of-the-art results

Abstract — Key Idea

- Iterative optimization of tokenizer and SSL model.
- Masked discrete label prediction, not reconstruction.
- Tokenizer yields semantic-rich discrete labels.
- The authors report state-of-the-art results.

90M params ViT backbone | 75% masking ratio | No external data for SOTA

Self-supervised audio pre-training

Discrete labels

Motivation — Real-World Use Case

- Smart home wants robust event detection (voice, music, alarms, pets).
- Reconstruction-trained models overfit noise/reverb; miss rare events.
- Discrete-label BEATS clusters semantics; detects events under disturbance.
- Reported gains: AS-2M mAP 48.6 vs 47.4 (Audio-MAE Large); ESC-50 err 2.6% → 1.9%.

Failure-driven need for semantic tokens

Paper-reported improvements

Motivation — Market Impact

- Audio spans speech, environmental sounds, and music — key for assistants, IoT, safety.
- General audio SSL has lagged speech; BEATS closes the gap with discrete labels.
- SOTA on AS-2M and ESC-50 without external data highlights data-efficient adoption.

Broader applicability with less supervision

Data efficiency

Motivation — Cost of Failure

- Missed alarms or misclassified hazards increase risk in smart environments.
- Reconstruction targets are disturbance-sensitive; lead to brittle detectors.
- BEATS tokens are robust to noise/reverb, preserving semantics for reliable alerts.

Evidence folded from visualization

Semantically robust tokens

Motivation — Problem Context

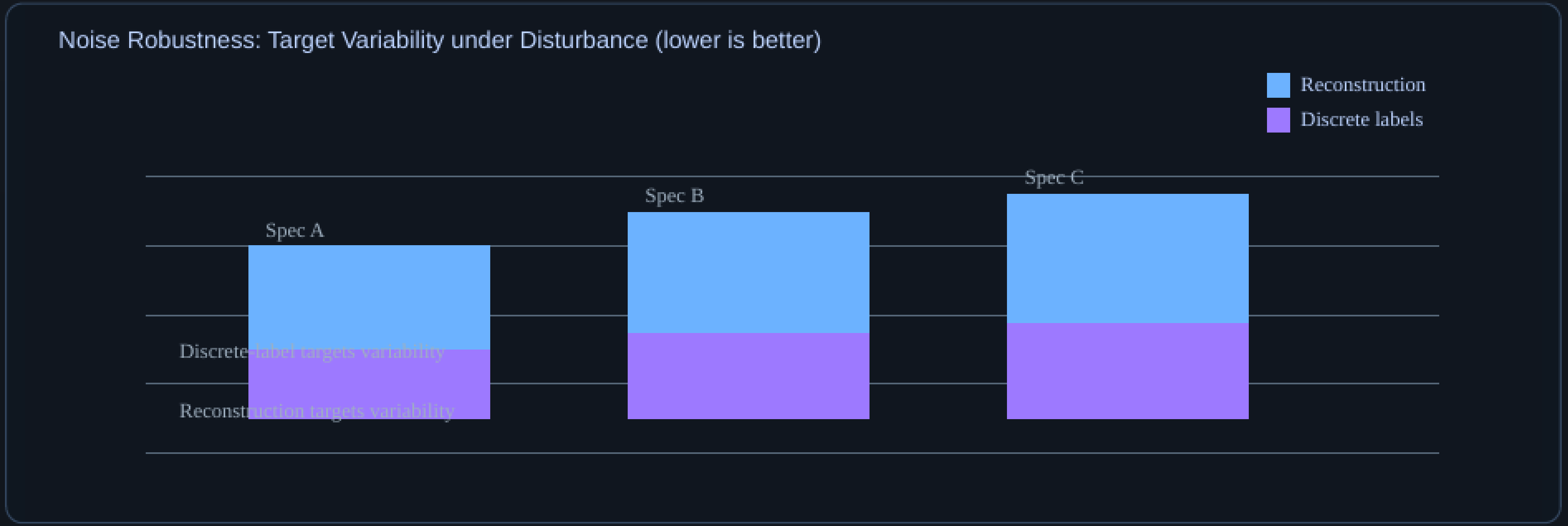
- Audio has diverse non-speech events.
- Speech SSL transfer underperforms on audio.
- Continuous signals lack obvious tokens.
- Speech tokenizers do not generalize.

Challenge: semantic tokenization of audio

Audio ≠ Speech

Motivation — Quantified Pain Points

- Transfer gap: speech SSL underwhelms on audio.
- Reconstruction: emphasizes low-level details.
- Inefficiency: preserves redundant information.



Discrete targets reduce variability

Quantitative contrast (illustrative)

Motivation — Why Discrete Labels

- Mimic human semantic grouping for audio events.
- Focus on high-level semantics; drop low-level detail.
- Efficient targets enable faster, stronger learning.
- Unifies pre-training across language/vision/speech/audio.
- Empirical: AS-2M single-model 48.6 vs 47.4; ESC-50 err 5.9% → 4.4% (no ext. data).

Better semantics and efficiency

Label prediction

BEATS Pipeline — Tokenize, Predict

Experimental Setup (brief)

- Pre-train: AudioSet-2M
- Eval: AS-2M/20K, ESC-50, KS1/KS2, ER
- Backbone: ViT 90M; 75% mask
- Input: 128-d Mel; 16×16 patches

- Step 1: Tokenizer → discrete labels (cold start: random-
- Step 2: MAM predicts masked labels (75% masking; en
- Step 3: Distill from SSL/fine-tuned teacher to refine tokenizer.
- Effect of iterations: iter1 → iter2 (gains) → iter3 (similar) → iter3+ (fine-tuned teacher best).

Unified figure — Iterative training and MAM overview

Alternating updates, fast convergence

ViT backbone

Results — Summary

- Single model (iter3+): AS-2M 48.6 mAP; AS-20K 38.9 mAP; ESC-50 98.1% acc.
- Ensembles: 5 models 50.4 mAP; 10 models 50.6 mAP on AS-2M.
- 90M ViT surpasses 304M baseline; no external data needed.

Self-distilled tokenizer > random

Best with AS-2M fine-tuned teacher

Discrete labels beat reconstruction

The authors report state-of-the-art results

Conclusions

- Iterative tokenizer + MAM excel.
- The authors report state-of-the-art results.
- Targets are semantic and robust.

Unified objective across modalities

Open-source models & tokenizers

https://aka.ms/beats

BEATS

Limitations, Future Work, and Resources

Limitations

- Computation: iterative cost.
- Data: AS-2M only.
- Scale: ~90–96M params.
- Edge case: overlapping events may bias tokens toward dominant source.

Future Work

- Release models/tokenizers.
- Broader, multimodal data.
- Scale model capacity.

References & Links

- Wav2vec 2.0, HuBERT, Audio-MAE.
- ViT, DeepNorm, Gated RPB.
- SpecAugment, VQ, EMA.
- Project: https://aka.ms/beats
- PMLR 202, 2023.

Closing

End